

Solution Requirements

Date	11 June 2026
Team ID	
Project Name	EcoTrack - Personal Carbon AI Footprint Tracker
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and

1. Functional Requirements Matrix:

- FR-1: Real-Time Emissions Calculator recording transport distance (km), diet type, food waste, and electricity (kWh).
- FR-2: Groq LLaMA 3.3 AI Recommendation Engine providing personalized lifestyle eco-swaps with CO2 savings.
- FR-3: 30-Day Historical Analytics Dashboard rendering Chart.js line trends, category breakdown charts, and heatmaps.
- FR-4: Gamification System computing logging streaks and unlocking 10 distinct achievement badges.
- FR-5: Firebase Firestore Data Synchronization & Firebase Auth User Authentication.

2. Non-Functional Requirements Matrix:

- NFR-1 (Performance): Aggregate backend API endpoints respond in < 180ms.
- NFR-2 (Availability): Application architecture ensures 99.9% uptime and availability.
- NFR-3 (Usability): ~~Responsive glassmorphism UI layout supporting desktop, tablet, and mobile viewports~~
- NFR-4 (Security): API keys (Groq, Carbon Interface, Firebase) secured via server-side .env environment variables.

1	Auth & Access Control	Secure user registration and authentication via Firebase Auth with JWT session tokens.	100% secure login & session persistence.
2	Data Collection	Wizard form capturing daily transport distance (km), diet choices, and utility electricity (kWh).	Multi-category emission calculation in < 100ms.
3	System Integration	Groq LLaMA 3.3 API gateway delivering personalized lifestyle eco-swaps with CO2 savings.	AI swap recommendations returned in < 850ms.

4	Transactions Processing	Real-time math calculations processing daily user inputs against emission factors.	Accurate CO2 emissions verified by Carbon Interface API.
5	Reporting & Dashboards	Chart.js visual dashboards rendering 30-day emission trends and category heatmaps.	Interactive chart rendering in < 150ms.
6	Business Rules & Logic	Streak counter tracking consecutive daily logs and unlocking 10 achievement badges.	Auto-unlock badges upon milestone completion.
7	Compliance & Privacy	GDPR-compliant data handling with user data encryption and privacy controls.	100% compliance with data privacy standards.
8	Cloud Persistence	Firebase Firestore cloud database synchronization with offline caching support.	Seamless offline-to-online database sync.

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	Fast REST API response times and rapid UI component rendering.	API response latency < 180ms; Page load < 1.2s.
2	Scalability	Serverless cloud hosting capable of scaling with user volume growth.	Supports up to 10,000 active concurrent users.
3	Security & Privacy	Encrypted client-server communications and secure environment variables.	HTTPS / TLS 1.3 encryption & zero plaintext keys.
4	Reliability & Availability	High availability application infrastructure with automated error recovery.	99.9% application uptime availability per month.
5	Usability & Accessibility	Responsive glassmorphism UI supporting desktop, tablet, and mobile viewports.	Complies with WCAG 2.1 AA accessibility guidelines.
6	Maintainability	Modular React component architecture and clean Flask REST API codebase.	100% modular code with full ESLint & PEP8 compliance.